Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from **3,900 purchases across various product categories**. The goal is to uncover insights into **spending patterns, customer segments, product preferences, and subscription behavior** to guide strategic business decisions.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18

Key Features:

- **Customer demographics** (Age, Gender, Location, Subscription Status)
- **Purchase details** (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- **Shopping behavior** (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- **Missing Data:** 37 values in **Review Rating** column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in **Python**:

Data Loading: Imported the dataset using **pandas**.

Initial Exploration: Used **df.info()** to check structure and **df.describe()** for summary statistics

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31

Previous Purchases	Payment Method	Frequency of Purchases
14	Venmo	Fortnightly
2	Cash	Fortnightly
23	Credit Card	Weekly
49	PayPal	Weekly
31	PayPal	Annually

Missing Data Handling: Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.

Column Standardization: Renamed columns to **snake case** for better readability and documentation.

Feature Engineering:

- Created **age_group** column by binning customer ages.
- Created **purchase_frequency_days** column from purchase data.

Data Consistency Check: Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.

Database Integration: Connected Python script to **PostgreSQL** and loaded the cleaned DataFrame into the database for SQL analysis.

Data Analysis using SQL

We performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90

3. Top 5 Products by Rating - Found products with the highest average review ratings.

item_purchased	average_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4. Average Purchase Amount by Shipping Type - Analyzed the average purchase amount for **Standard** and **Express** shipping methods. This helps understand whether customers choosing faster delivery tend to spend more.

item_purchased	average_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

5. Spending Behavior of Subscribed vs Non-Subscribed Customers - Analyzed whether **subscribed customers spend more** compared to non-subscribers. Compared the **average purchase amount and total revenue** generated by both groups

subscription_status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

6. Top 5 Products with Highest Discount Usage - Identified the products with the **highest percentage of purchases where discounts were applied**. This helps understand which products are most frequently bought using discounts.

item_purchased	discount_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation Based on Purchase History - Segmented customers into New, Returning, and Loyal based on their previous purchases. This helps understand the distribution of customers across different loyalty levels.

customer_segment	number of customers
returning	1478
loyal	2339
new	83

8. Top 3 Most Purchased Products in Each Category - Identified the **top 3 most purchased products within each category** based on total orders. This helps understand which products are the most popular in each product category.

customer_segment	number of customers
returning	1478
loyal	2339
new	83

9. Repeat Buyers vs Subscription Status - Analyzed whether **repeat buyers are more likely to be subscribed customers**. This helps understand the relationship between **customer loyalty and subscription status**.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145

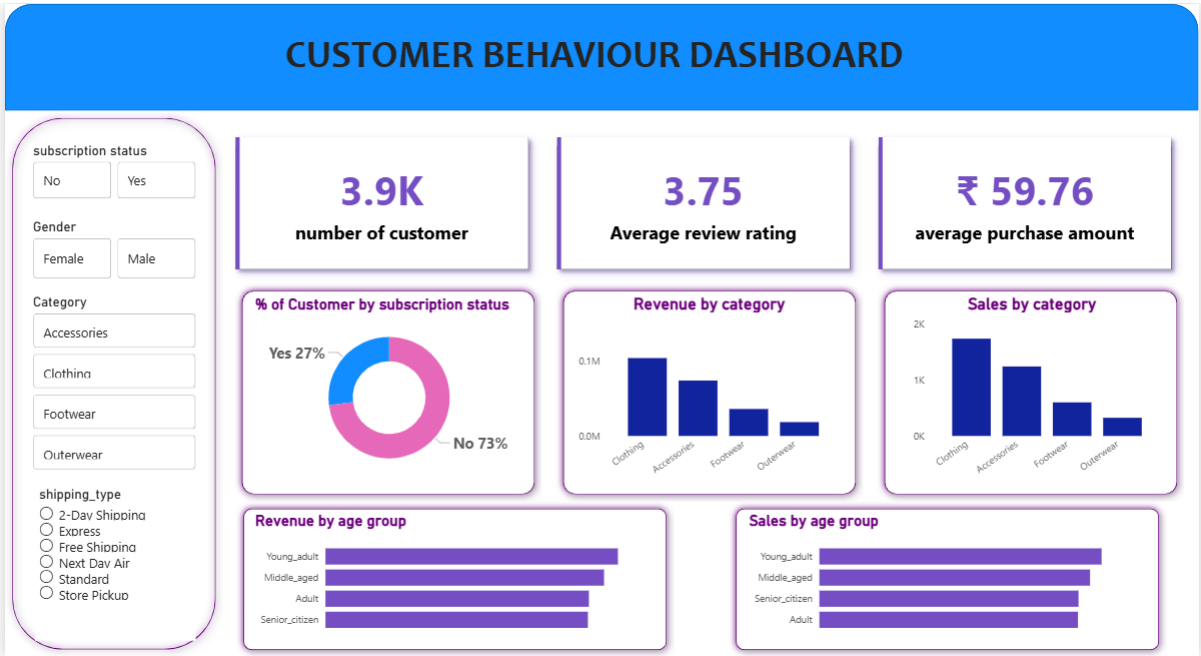
10. Revenue Contribution by Age Group - Analyzed the **total revenue generated by each age group**. This helps identify which age group contributes the most to overall sales.

subscription_status	repeat_buyers
Yes	958
No	2518

The **Customer Behaviour Dashboard** is created using Microsoft Power BI to analyze customer purchasing patterns and business performance. It provides a visual representation of important customer data such as total customers, revenue, sales, and customer ratings. The dashboard helps organizations understand customer preferences and make better business decisions.

Key points of the dashboard:

- Displays **total number of customers (3.9K)** to show the size of the customer base.
- Shows **average review rating (3.75)** which represents overall customer satisfaction.
- Indicates **average purchase amount (₹59.76)** spent by customers.
- A **donut chart** presents the percentage of customers by subscription status, where **73% are non-subscribers and 27% are subscribers**.
- **Revenue by category** visualization shows that **Clothing generates the highest revenue**, followed by Accessories, Footwear, and Outerwear.
- **Sales by category** highlights that Clothing products have the highest number of sales.
- **Revenue by age group** shows that **Young Adults contribute the highest revenue** compared to other age groups.
- **Sales by age group** indicates that Young Adults make the most purchases.
- Interactive **filters (slicers)** allow users to analyze data based on **subscription status, gender, category, and shipping type**.
- The dashboard helps businesses **analyze trends, understand customer behavior, and improve sales strategies**



Business Decisions ->

- **Focus on Clothing Category:** Since clothing generates the highest revenue and sales, the company should increase inventory and marketing efforts for clothing products.

- **Increase Customer Subscriptions:** As only **27% customers are subscribed**, the business can introduce special discounts, loyalty programs, or membership benefits to increase subscriptions.

- **Target Young Adult Customers:** Young adults contribute the highest revenue and sales, so marketing campaigns and advertisements should focus more on this age group.

- **Improve Low Performing Categories:** Categories like **Footwear and Outerwear** generate lower revenue, so the company can improve product variety, quality, or promotions to increase their sales.

- **Enhance Customer Satisfaction:** The **average rating of 3.75** indicates moderate satisfaction. Businesses should improve product quality and customer service to increase ratings.

- **Optimize Shipping Options:** By analyzing shipping preferences, companies can focus on the most used shipping methods and improve delivery services.

- **Personalized Marketing:** Using filters such as **gender, category, and subscription status**, businesses can create personalized offers and targeted promotions for different customer segments.

Thank You